

SACRAMENTO EXECUTIVE, CA, USA

WMO: 724830

Lat: 38.507N

Lon: 121.495W

Elev: 5

StdP: 101.27

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 23232

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-0.3	1.1	-6.5	2.2	11.8	-4.4	2.6	10.7	11.1	12.4	9.5	11.1	0.5	330	0.463

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.7	37.9	20.9	36.1	20.2	34.3	19.6	22.3	35.1	21.3	33.6	20.4	32.2	3.1	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.5	12.6	28.4	16.4	11.7	26.1	15.6	11.1	24.9	65.1	34.9	61.4	33.8	58.4	32.2	27.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8.9	7.8	6.8	-2.4	41.2	1.5	1.4	-3.6	42.1	-4.4	42.9	-5.3	43.7	-6.4	44.7
			-3.1	23.5	1.6	1.3	-4.2	24.4	-5.1	25.2	-6.0	25.9	-7.2	26.8
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	16.5	8.8	10.7	13.0	15.2	18.7	22.2	24.3	23.9	22.3	17.8	12.2	8.6	(6)
	DBStd	6.32	2.58	2.52	2.88	3.33	3.43	3.35	2.86	2.64	2.84	2.92	2.98	2.81	(7)
	HDD10.0	154	53	20	7	2	0	0	0	0	0	0	13	61	(8)
	HDD18.3	1353	294	214	165	106	36	5	0	0	3	45	186	301	(9)
	CDD10.0	2531	17	39	101	157	271	367	442	432	368	242	78	18	(10)
	CDD18.3	688	0	0	1	11	48	122	184	173	121	28	1	0	(11)
	CDH23.3	8581	0	1	34	202	709	1576	2207	2006	1417	414	16	0	(12)
	CDH26.7	4386	0	0	2	59	307	819	1243	1103	717	135	1	0	(13)
(14) Wind	WSAvg	2.7	2.2	2.6	2.8	3.1	3.3	3.3	3.2	2.9	2.5	2.3	1.9	2.3	(14)
(15) Precipitation	PrecAvg	504	95	90	73	36	21	6	0	1	2	27	51	102	(15)
	PrecMax	833	242	277	221	113	94	37	4	12	12	99	116	249	(16)
	PrecMin	197	1	0	2	3	0	0	0	0	0	4	5	4	(17)
	PrecStd	140	73	62	63	26	22	10	1	3	4	23	30	66	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	18.9	22.1	26.4	31.4	35.9	39.4	40.1	40.0	37.9	33.5	25.4	18.1	(19)
		MCWB	11.3	14.1	16.3	17.6	20.1	21.5	22.0	21.2	19.8	18.2	15.3	12.7	(20)
	2%	DB	16.6	19.7	24.0	28.3	32.9	36.6	37.7	37.2	35.6	30.1	22.7	16.2	(21)
		MCWB	11.7	13.3	15.2	16.2	18.9	20.5	21.3	20.6	19.1	16.9	14.4	12.2	(22)
	5%	DB	14.9	17.7	21.9	25.6	30.3	34.2	35.6	35.0	33.4	27.7	20.1	14.9	(23)
		MCWB	11.4	12.2	14.3	15.7	17.3	19.4	20.4	19.9	18.5	15.7	13.4	11.6	(24)
	10%	DB	13.4	15.9	19.7	22.8	27.4	31.4	33.4	32.8	31.1	25.4	17.9	13.7	(25)
		MCWB	10.7	11.6	13.3	14.4	16.5	18.8	20.0	19.4	18.0	15.1	12.6	10.7	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.7	15.8	17.7	19.5	21.5	23.0	24.1	24.0	22.0	19.6	16.9	15.1	(27)
		MCDB	16.1	19.1	24.4	27.5	33.0	36.4	36.9	36.2	33.6	29.6	22.8	16.8	(28)
	2%	WB	13.1	14.6	16.3	17.7	19.8	21.6	22.3	21.7	20.3	18.1	15.6	13.8	(29)
		MCDB	14.8	17.9	22.5	26.0	31.1	34.3	35.3	34.6	31.7	27.5	20.3	15.2	(30)
	5%	WB	12.0	13.4	15.1	16.4	18.5	20.4	21.3	20.6	19.4	17.0	14.6	12.5	(31)
		MCDB	13.5	16.3	20.4	24.1	28.6	32.2	33.7	32.9	30.9	25.2	18.5	13.9	(32)
	10%	WB	11.2	12.3	13.9	15.2	17.2	19.3	20.4	19.8	18.6	16.0	13.5	11.4	(33)
		MCDB	12.9	14.9	18.5	21.6	25.9	30.2	32.1	31.5	29.8	23.1	16.8	13.0	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	8.5	9.9	11.5	12.9	14.7	16.5	17.7	17.5	16.8	14.6	11.3	8.9	(35)
		MCDBR	11.4	13.4	15.5	17.6	19.2	20.3	20.2	20.4	20.3	18.5	14.9	10.7	(36)
		MCWBR	7.0	7.8	8.1	7.8	7.3	7.2	7.3	7.0	6.6	7.0	7.4	6.6	(37)
	5% WB	MCDBR	7.6	10.7	13.8	16.0	18.1	19.0	19.2	18.8	18.1	15.9	12.0	6.4	(38)
		MCWBR	5.2	6.8	7.7	7.9	7.7	7.6	7.7	7.1	6.6	6.5	6.7	4.6	(39)
(40) Clear-Sky Solar Irradiance	taub		0.324	0.326	0.341	0.356	0.365	0.362	0.363	0.347	0.351	0.345	0.331	0.333	(40)
	taud		2.503	2.504	2.449	2.399	2.395	2.413	2.407	2.442	2.437	2.463	2.486	2.473	(41)
	Ebn at Noon		855	903	920	922	914	914	910	919	896	866	837	813	(42)
	Edh at Noon		79	89	103	116	118	116	116	109	103	91	79	76	(43)
(44) All-Sky Solar Radiation	RadAvg		2.16	3.23	4.62	6.11	7.41	8.25	8.21	7.35	5.96	4.22	2.70	1.94	(44)
	RadStd		0.41	0.40	0.47	0.51	0.43	0.32	0.23	0.23	0.20	0.27	0.19	0.32	(45)

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	-0.52	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (48 neighbors)	+0.40	N/A	-1.08	N/A	N/A	N/A	N/A	-116	+138	+42	(47)

Nomenclature: See separate page